

Quantum Computing: Coming Faster Than You Think

From cybersecurity to drug discovery, there are several challenges that quantum computing can tackle better than classical systems. With improving hardware, full-stack solutions, better algorithms, simulators, and cloud offerings, the technology is at present in ‘democratisation’ phase. Perhaps it is time to make your quantum plans too.

IBM Quantum Lab at the Thomas J. Watson Research Center (Credit: Connie Zhou for IBM)

“Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city?” First formulated in 1930, this mathematical problem, known as the ‘travelling salesman problem,’ is one of the hardest to solve, yet one of the most essential optimisation issues for any logistics provider.

“Quantum computing has the potential to blow classical computing out of the water for certain types of complicated problems. For example, constrained optimisation challenges like the famous travelling salesman problem can have trillions of equations and variables. The most complex of these problems will simply be unfeasible for classical computers to tackle, but quantum computers will be able to solve them

in seconds. And these aren’t just laboratory problems; constrained optimisation challenges heavily impact industries like supply chain and retail, cybersecurity, life sciences and pharmaceutical, and government and applied research,” says Robert Liscouski, CEO of Quantum Computing Inc. (QCI).

Another great example involves autonomous vehicles. QCI recently solved a 3854-variable sensor placement problem for BMW using only quantum computing—and they did it in six minutes. This is a clear example of quantum computing’s ability to solve actionable, real-world problems that would take classical computers a much longer time to solve.

“To clarify, sensor placement is extremely challenging because of all the variables that must be considered, including wind resistance, weight

